

Instruction-Conditioned Electromagnetic Spectrum Understanding via Budget-Adaptive Signal Tokenization

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Abstract

Electromagnetic spectrum monitoring increasingly requires flexible analysis beyond task-specific recognition and detection. Multimodal large language models offer a unified interface, but extending vision-language models (VLMs) to raw I/Q signals requires tokenization that balances fidelity against a strict budget. For signals, dense encoding causes token costs to grow with observation length, whereas fixed-resolution compression may discard short-duration or localized signal evidence. Thus, we propose **BATok**, a budget-adaptive signal tokenizer that adjusts token capacity to the input length while allocating that capacity according to the signal content. BATok constructs candidate representations from signal-derived features using lightweight multi-resolution branches, then combines a local energy prior with learnable queries to resample these representations into compact signal tokens. The number of tokens adapts to the input length while remaining strictly bounded. The resulting tokens are projected into the language embedding space of VLMs. We further introduce **EMSpec-Instruct**, a multimodal instruction dataset aligning raw I/Q signals, waterfall images, and language supervision for modulation recognition, structured detection, and language-conditioned signal grounding. Experiments show that BATok learns effective signal representations and achieves competitive performance across all tasks.

Introduction

Electromagnetic spectrum monitoring supports communications, spectrum management, interference analysis, and dynamic spectrum access (Yucek and Arslan 2009; Ding et al. 2018). Beyond recognizing predefined signal classes, practical monitoring also requires detecting, localizing, and characterizing signals of interest across time and frequency (Ding et al. 2018; Boegner et al. 2022b). These practical demands call for spectrum analysis systems that can flexibly respond to different information needs rather than producing only a fixed set of predictions. Figure 1 contrasts this unified objective with task-specific pipelines.

Existing spectrum-analysis methods generally use task-specific encoders with dedicated classification or detection heads (O’Shea, Corgan, and Clancy 2016; Shalu and Singh 2026; Boegner et al. 2022b). Although effective for individual objectives, they expose separate interfaces for recognition, detection, and localization. Supporting additional

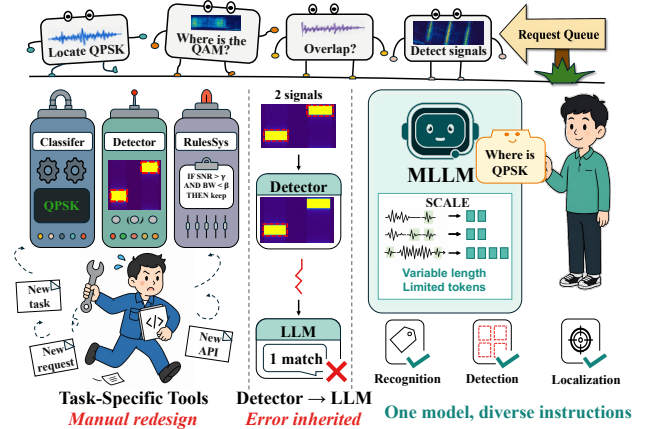


Figure 1: Motivation for unified instruction-conditioned spectrum understanding. Task-specific models require manual redesign, while detector-LLM cascades inherit upstream errors. Our framework couples signal tokenization with a VLM through a unified interface.

queries therefore typically requires manually designed rules, post-processing, or new model components. A detector-LLM cascade can provide a natural-language interface, but its answers remain constrained by the labels, boxes, and attributes produced by the upstream detector. In contrast, multimodal large language models (MLLMs) provide a general interface for multimodal perception, reasoning, and language generation (Yin et al. 2024; Jiang et al. 2026).

Building on recent advances in audio-language models (KimiTeam et al. 2025; Yang, Ho, and Lee 2025), video-language models (Zhang et al. 2025; Tang et al. 2026), and time-series-language models (Wang et al. 2025; Liu et al. 2025a), recent RF-oriented MLLMs connect electromagnetic observations to language through visual representations, including waveform plots, spectra, spectrograms, waterfall images, and constellation diagrams (Zou et al. 2026b,a; Han et al. 2026; Bara et al. 2026). However, these vision-aligned interfaces convert numerical I/Q samples into fixed-resolution images before tokenization. Such conversion may attenuate waveform-level amplitude, phase, and transient evidence, while the visual token capacity is de-

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terminated by image resolution rather than the duration and content of the original observation. Moreover, generic audio and time-series tokenizers are not designed around the coupled physical structure of I/Q samples.

Existing signal encoders achieve strong performance on task-specific signal recognition using CNN-based (Du et al. 2022), RNN-based (Yao, Yang, and Huang 2025), Transformer-based (Zhai et al. 2025), and state-space-based (Zhang et al. 2024; Ma, Wang, and Zhang 2025) architectures. Nevertheless, they typically produce global representations for fixed prediction heads or temporal features whose granularity is determined by predefined strides, pooling ratios, or patch sizes. This creates a capacity mismatch for mixed-length observations: dense representations become expensive for long signals, whereas a uniformly small output budget may discard localized evidence. Consequently, existing encoders generally do not adapt their output granularity jointly to input length and information distribution. Integrating raw I/Q into an MLLM therefore requires a tokenizer that adapts its representation capacity to signal length and content while keeping the output strictly bounded.

To address these challenges, we develop an instruction-conditioned multimodal framework for electromagnetic spectrum understanding. At its core, we introduce **BATok**, a signal tokenizer that adapts the total token capacity to the input length and adapts temporal aggregation to the signal content for VLM conditioning. BATok first constructs candidate signal representations using signal-derived amplitude, power, and phase features together with lightweight multi-resolution temporal branches. It performs evidence-guided query resampling to compress variable-length features into compact signal tokens, while a monotonic token-budget function allocates larger yet strictly bounded representation capacity to longer observations. The resulting tokens are projected into the language-model embedding space and jointly processed with waterfall-image tokens and textual instructions. To provide supervision for this setting, we construct **EMSpec-Instruct**, a multimodal instruction dataset that aligns raw I/Q observations and waterfall images with modulation recognition, structured detection, and language-conditioned signal grounding.

Our main contributions are summarized as follows:

- We construct **EMSpec-Instruct**, a multimodal instruction dataset aligning I/Q signals, waterfall images, and language supervision for modulation recognition, structured detection, and language-conditioned signal grounding.
- We propose **BATok**, a lightweight signal tokenizer with a compact parameter footprint. It uses multi-resolution evidence-guided query resampling to convert mixed-length I/Q observations into informative tokens with an adaptive yet strictly bounded budget.
- We integrate BATok with a VLM through continuous signal-token projection. Experiments demonstrate the effectiveness and show that I/Q signals provide complementary waveform-level evidence beyond waterfall images.

Related Work

Representation Learning from Raw I/Q Signals

Deep automatic modulation classification established that useful radio representations can be learned directly from complex I/Q sequences. CNN-based methods learn local temporal and cross-channel patterns (O’Shea, Corgan, and Clancy 2016; Xu et al. 2020); DNCNet further couples convolutional denoising and classification for robust recognition (Du et al. 2022). Transformer-based models capture long-range dependencies, as represented by SiT’s dual-domain interaction and IQFormer’s multi-representation sequence modeling (Zhai et al. 2025; Shao et al. 2025). More recently, MAMC and DWMTN employ selective state-space modeling, alone or with Transformers, for efficient long-sequence processing (Zhang et al. 2024; Ma, Wang, and Zhang 2025); self-supervised methods such as RIS-MAE learn transferable features through masked reconstruction (Liu et al. 2025b). A broader survey is provided in (Shalu and Singh 2026). These architectures demonstrate the value of waveform-level amplitude, phase, and temporal evidence, but predominantly produce global or fixed-granularity features for task-specific heads. BATok instead converts mixed-length I/Q observations into a bounded continuous-token sequence suitable for multimodal language models.

Connecting Non-Textual Sequences to Language Models

Non-textual observations are commonly connected to LLMs through patch-based or query-based interfaces. Audio and video MLLMs use learned cross-modal connectors (Tang et al. 2024; Zhang, Li, and Bing 2023), while Time-LLM, ChatTime, and ChatTS introduce numerical sequence representations for forecasting, understanding, and reasoning (Jin et al. 2024; Wang et al. 2025; Xie et al. 2025). These methods establish the feasibility of continuous inputs, but generic sequence interfaces do not explicitly preserve coupled I/Q evolution; moreover, fixed-size patches yield token counts that grow with duration. Query-based resamplers instead map variable encoder features to a fixed token set, as in Flamingo and BLIP-2 (Alayrac et al. 2022; Li et al. 2023). This bounds the interface but assigns equal capacity to observations of different lengths. BATok retains a bounded interface while adapting token count to valid length and allocating that budget using multi-resolution features, learned relevance, and an energy prior.

RF-Oriented Language and Multimodal Models

Signal-oriented language and multimodal models follow two main directions. Numerical approaches adapt language-model backbones to radio signal tasks: RadioLLM reprograms I/Q patches and combines language and convolutional features for classification and reconstruction, retaining a representation-plus-task-head formulation (Chen et al. 2025); Large Spectrum Models discretize spectrum measurements for decoder-only forecasting (Lunar and Vuran 2026). Visual approaches reuse VLM interfaces by rendering RF observations. Seeing Radio uses I/Q traces and spectrograms for modulation VQA (Zou et al. 2026b); RF-GPT, PRoD,

and RF-Analyzer extend this paradigm to broader spectrum understanding, electromagnetic tasks, and synthetic-to-real waterfall analysis (Zou et al. 2026a; Han et al. 2026; Bara et al. 2026). Although effective for language-conditioned RF analysis, their token layout and capacity are inherited from the rendered image grid rather than the length and information distribution of numerical I/Q observations.

BATok: Budget-Adaptive Signal Tokenization

Integrating raw I/Q signals into a VLM requires more than extracting waveform features. The representation must allocate sufficient capacity to observations of different lengths, preserve signal information that is distributed non-uniformly over time, and remain compatible with the bounded context of the language model. We address this problem at two levels: a length-dependent token budget controls the overall representation capacity, while signal-aware resampling determines how that capacity is distributed over the temporal observation. Figure 2 summarizes the complete BATok pipeline and its integration with image and text tokens.

We propose **BATok**, which adapts the total token capacity to the input length and adapts temporal aggregation to the content for VLM conditioning.

Given a raw I/Q sequence $X = \{x_t\}_{t=1}^T$ with $x_t = I_t + jQ_t$, BATok learns the mapping

$$f_\theta : \mathbb{R}^{T \times 2} \rightarrow \mathbb{R}^{K(T) \times d_s}, \quad K(T) \leq K_{\max}, \quad (1)$$

where T is the number of valid input samples, $K(T)$ is a length-dependent token budget, $K_{\max}$ is the maximum signal-token budget allowed by the VLM context, and d_s is the signal-token dimension. The budget function $K(T)$ controls the global representation capacity, while signal-aware resampling determines how that capacity is distributed over the observation.

BATok first augments the raw I/Q sequence with signal-derived features and constructs candidate temporal representations at complementary resolutions. It then combines learned feature relevance with a local energy prior to modulate the branch representations and uses learnable queries to resample them into a compact set of scale-specific tokens. The resulting tokens are tagged by their temporal scale and assembled into the final signal representation.

Signal-Derived Feature Enhancement

Raw I/Q samples provide the Cartesian components of a complex baseband signal, with amplitude and phase implicitly determined by their joint variation. Although a sufficiently expressive network can in principle recover these quantities from I_t and Q_t , requiring a lightweight tokenizer to learn the corresponding nonlinear transformations may unnecessarily consume model capacity. We therefore retain the I/Q components and augment them with signal-derived features that expose complementary views of the underlying waveform.

For each sample, we construct

$$u_t = [I_t, Q_t, A_t, P_t, \Delta\phi_t], \quad (2)$$

where $A_t = \sqrt{I_t^2 + Q_t^2}$, $P_t = I_t^2 + Q_t^2$, and $\Delta\phi_t = \angle(x_t x_{t-1}^*)$ denote the instantaneous amplitude, power, and

inter-sample phase difference, providing complementary descriptions of the signal. These derived features provide complementary descriptions of envelope variation, relative signal activity, and local phase evolution. Although A_t and P_t are deterministically related, their different nonlinear scales expose waveform variations from complementary perspectives.

The enhancement does not replace or discretize the original waveform; instead, it preserves I_t and Q_t while providing explicit signal-domain coordinates for envelope, energy, and phase evolution. The resulting sequence $U = \{u_t\}_{t=1}^T$ is projected into a shared hidden space by a lightweight convolutional stem:

$$H_0 = \text{GELU}(\text{GN}(\text{Conv1D}(U))), \quad (3)$$

where GN denotes Group Normalization (Wu and He 2018) and $H_0 \in \mathbb{R}^{T \times d_h}$ is passed to the subsequent multi-resolution temporal encoder. In the VLM experiments, $d_h = d_s = 256$.

Multi-Resolution Temporal Encoding

Electromagnetic observations may contain informative structures over substantially different temporal extents. Fine temporal resolution helps preserve localized waveform variations, whereas coarser representations provide broader temporal coverage and stronger sequence compression. A single receptive field and downsampling ratio may offer an inadequate trade-off between local detail and contextual coverage. BATok instead constructs complementary temporal representations at multiple resolutions before adaptive resampling.

Let $\mathcal{B}$ denote the set of temporal branches operating at different resolutions. Each branch applies a strided depthwise-separable one-dimensional convolution:

$$H_b = \text{DSConv}_b(H_0) \in \mathbb{R}^{T_b \times d_h}, \quad b \in \mathcal{B}, \quad (4)$$

where T_b is the effective length after downsampling of branch b . The receptive field and downsampling ratio are set to increase progressively across the branches, producing different trade-offs among temporal detail, contextual coverage, and representation length. The fine-resolution branch retains localized waveform variations, the intermediate-resolution branch captures structures over moderate temporal extents, and the coarse-resolution branch summarizes broader contexts under stronger compression.

After downsampling, each branch is refined by the lightweight temporal and channel mixing without changing its sequence length:

$$\begin{aligned} \tilde{H}_b &= H_b + \text{DWConv}_b(\text{LN}(H_b)), \\ H'_b &= \tilde{H}_b + \text{GLU}\left(\text{PWConv}_b\left(\text{LN}(\tilde{H}_b)\right)\right). \end{aligned} \quad (5)$$

Here, LN denotes Layer Normalization. DWConv and PWConv follow the depthwise and pointwise operations in depthwise separable convolution (Howard et al. 2017), and GLU denotes the gated linear unit (Dauphin et al. 2017). This factorized design enriches each branch representation with linear complexity in the sequence length, avoiding the quadratic cost of global self-attention (Vaswani et al. 2017). The feature sequences $\{H'_b\}_{b \in \mathcal{B}}$ form a multi-resolution representation pool for the subsequent adaptive resampling stage, rather than being directly forwarded to the VLM.

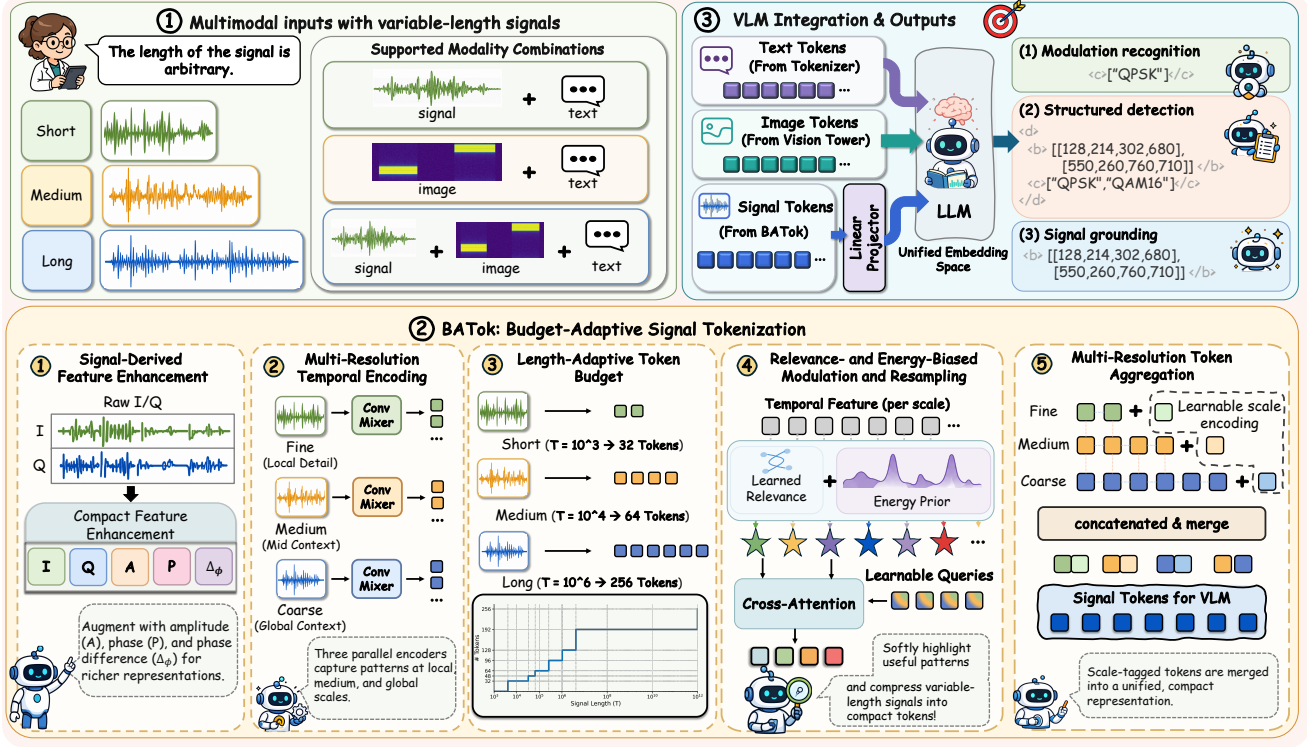


Figure 2: Overview of BATok and its integration with a VLM. Variable-length I/Q signals are augmented with signal-derived features, encoded at multiple temporal resolutions, assigned a bounded length-adaptive token budget, and resampled using learned relevance and an energy prior. The resulting multi-resolution signal tokens are aggregated, projected into the language embedding space, and jointly processed with image and text tokens for unified spectrum understanding.

Adaptive Token Budgeting and Resampling

The multi-resolution encoder produces branch features whose temporal lengths still vary with the input length. BATok controls the overall representation capacity through a length-dependent but strictly bounded token budget and subsequently allocates this capacity over time through signal-aware query resampling.

Length-adaptive token budget. A fixed token budget imposes increasingly severe compression as the signal length grows, which may preserve redundant capacity for short inputs while discarding more temporal detail from long inputs. Conversely, assigning tokens in direct proportion to the input length causes the representation cost to grow continuously with T and may exceed the bounded context available to the VLM. Therefore, the total number of signal tokens is determined by a monotonic piecewise-constant function:

$$K(T) = \text{clip} \left(K_{\min} + \sum_{j=1}^J \Delta_j \mathbb{I}(T > \tau_j), K_{\min}, K_{\max} \right), \quad (6)$$

where τ_j are increasing length thresholds and $\Delta_j \geq 0$ are budget increments. $\mathbb{I}(\cdot)$ denotes the indicator function. Consequently, longer signals are assigned more tokens, while the resulting token sequence remains compatible with the bounded context of the VLM. The piecewise-constant sched-

ule also assigns inputs within the same threshold interval to use the same token budget, avoiding unnecessary changes in token count for small variations.

The total budget is distributed across the resolution branches according to predefined proportions ρ_b , where $\rho_b \geq 0$ and $\sum_{b \in \mathcal{B}} \rho_b = 1$. Integer branch budgets $K_b = \lfloor \rho_b K(T) \rfloor$ are obtained from these proportions and adjusted to preserve the total token count $\sum_{b \in \mathcal{B}} K_b = K(T)$.

Relevance- and Energy-Biased Modulation. Although all valid temporal positions are retained, they may contribute differently to the compact representation. A learned relevance score captures task-dependent information from the latent features, while local energy provides a complementary indication of temporal signal activity. However, high-energy regions may also arise from interference, impulsive noise, or unrelated transmissions. BATok therefore uses energy only as a soft bias rather than as a direct criterion for selecting temporal positions.

For the t -th feature $h'_{b,t}$ of branch b , BATok combines a learned relevance score with a local energy prior:

$$\begin{aligned} E_{b,t} &= \text{AvgPool}_{k_b, s_b}(I^2 + Q^2)_t \\ r_{b,t} &= \text{MLP}(h'_{b,t}) + \lambda_b e_{b,t}, \end{aligned} \quad (7)$$

where k_b and s_b denote the pooling kernel size and stride aligned with the temporal resolution of branch b . $e_{b,t}$ is

the normalization of $E_{b,t}$. The term $\text{MLP}(h'_{b,t})$ denotes the learned relevance score, while λ_b is a learnable coefficient controlling the contribution of the normalized energy prior.

The combined scores are normalized over valid temporal positions and used to modulate the branch features:

$$\bar{H}_b = H'_b \odot \left[1 + \text{clip} \left(\sqrt{T_b} \text{softmax}(r_b), 0, c \right) \right], \quad (8)$$

where c bounds the maximum enhancement magnitude, and $\odot$ denotes element-wise multiplication with broadcasting over the channel dimension. The factor $\sqrt{T_b}$ compensates for the reduced magnitude of normalized scores in longer sequences, while clipping prevents a small number of positions from being amplified. This modulation does not remove low-scoring positions; instead, it introduces a soft bias before query-based resampling, allowing the subsequent cross-attention module to consider the complete valid sequence.

Query-based resampling. Learnable-query interfaces have been used to compress variable-size perceptual features before language-model conditioning (Alayrac et al. 2022; Li et al. 2023). BATok adapts this interface to raw I/Q by assigning a length-dependent number of queries to each temporal scale. Each modulated branch is converted into a bounded set of tokens through cross-attention with learnable queries and then refined by a lightweight residual feed-forward layer:

$$\begin{aligned} Z_b &= \text{CrossAttn} \left(Q_b^{(K_b)}, \bar{H}_b, \bar{H}_b \right), \\ \hat{Z}_b &= Z_b + \text{FFN}(\text{LN}(Z_b)), \end{aligned} \quad (9)$$

where $Q_b^{(K_b)} = \{q_{b,i}\}_{i=1}^{K_b} \in \mathbb{R}^{K_b \times d_h}$ contains the first K_b queries from the learnable query pool of branch b . This process compresses each variable-length branch representation into K_b scale tokens, keeping the signal-token length adaptive yet bounded while emphasizing active signal regions and locally informative temporal structures.

Multi-Resolution Token Aggregation

The scale-tagged tokens are concatenated and refined by a shared token-wise residual projection. Explicit cross-scale interactions are subsequently modeled by the VLM. Since these tokens are extracted at different temporal resolutions, they contain distinct scale-specific biases. We therefore add a learnable scale encoding $\eta_b \in \mathbb{R}^{d_s}$ to each branch:

$$\tilde{Z}_b = \hat{Z}_b + \eta_b, \quad b \in \mathcal{B}. \quad (10)$$

The scale-aware tokens are concatenated and passed through a shared token-wise residual projection:

$$Z = \text{Concat}_{b \in \mathcal{B}}(\tilde{Z}_b), \quad (11)$$

$$Z_{\text{sig}} = Z + \text{GELU}(\text{Linear}(\text{LN}(Z))). \quad (12)$$

where $Z \in \mathbb{R}^{K(T) \times d_s}$. The final representation $Z_{\text{sig}} \in \mathbb{R}^{K(T) \times d_s}$ serves as compact multi-resolution signal tokens, which are further projected into the VLM language embedding space.

VLM Integration and Training

Following common practice in multimodal large language models (Alayrac et al. 2022; Li et al. 2023; Liu et al. 2024), the signal tokens are not discretized into textual symbols, but are mapped into the VLM language embedding space as continuous multimodal tokens. After obtaining compact signal tokens $Z_{\text{sig}} \in \mathbb{R}^{K(T) \times d_s}$ from BATok, we project them into the VLM embedding space using a two-layer projector:

$$H_s = W_2 \text{GELU}(W_1 \text{LN}(Z_{\text{sig}})) \in \mathbb{R}^{K(T) \times d_\ell}, \quad (13)$$

where W_1 maps the signal-token dimension d_s to the VLM language embedding dimension d_ℓ , and W_2 preserves d_ℓ . The projected embeddings H_s are fed into the VLM together with available image embeddings and textual instruction embeddings. Image embeddings are obtained from the original visual encoder of the VLM, while text embeddings are produced by its token embedding layer. Therefore, the model can support three input configurations: image-text, signal-text, and image-signal-text inputs.

We train the integrated model with standard supervised fine-tuning. Given $\mathcal{D} = \{(X^{(n)}, V^{(n)}, P^{(n)}, Y^{(n)})\}_{n=1}^N$, where $X^{(n)}$ and $V^{(n)}$ denote optional signal and waterfall-image inputs, $P^{(n)}$ is the textual instruction, and $Y^{(n)}$ is the target response, we optimize the response-only autoregressive objective:

$$\mathcal{L}_{\text{sft}} = - \sum_{n=1}^N \sum_{i=1}^{|Y^{(n)}|} \log p_\phi \left(y_i^{(n)} \mid X^{(n)}, V^{(n)}, P^{(n)}, y_{<i}^{(n)} \right). \quad (14)$$

The loss is computed only on response tokens, while image, signal, and instruction tokens serve as conditioning context.

EMSpec-Instruct Dataset

Existing electromagnetic datasets are primarily designed for task-specific recognition or detection, typically providing class labels for individual I/Q segments or bounding-box annotations on time-frequency representations (Tekbıyık et al. 2020; Boegner et al. 2022a,b). Such supervision does not directly support instruction-conditioned recognition, structured multi-signal detection, and language-conditioned signal grounding. We therefore construct **EMSpec-Instruct**, a multimodal instruction dataset that associates raw I/Q observations, waterfall images, structured signal metadata, and natural-language queries.

The dataset integrates both single-signal and multi-signal observations. The single-signal stream combines HISARMOD-SUB, a subset of HisarMod2019.1 (hereafter HisarMod2019) (Tekbıyık et al. 2020), and EMSPEC-SIM-SUB, a subset of our single-signal simulation benchmark EMSPEC-SIM, for modulation recognition. The multi-signal stream combines SpaceNet (ChaSpark 2026) and a multi-signal simulated dataset constructed with RF simulation utilities (Boegner et al. 2022a), supporting structured detection and signal grounding. The simulated data complement the real datasets by expanding modulation coverage and introducing diverse acquisition lengths, signal densities, and time-frequency overlap patterns.

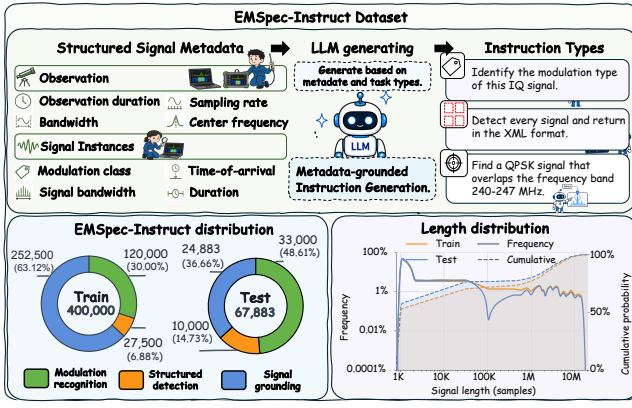


Figure 3: Construction and statistics of EMSpec-Instruct. Structured observation- and instance-level metadata are converted into metadata-grounded instructions. The bottom panels summarize the task composition of the training and test sets and the distribution of signal lengths.

To associate electromagnetic observations with language supervision, we first derive task conditions from the structured metadata and then use Qwen3.5-Max (Team 2026) to generate linguistically diverse instruction records. For modulation recognition and structured detection, Qwen3.5-Max generates the question while the compact answer is filled directly from the class label or complete scene annotation. For language-conditioned signal grounding, Qwen3.5-Max generates both the metadata-conditioned question and answer; the result is retained only when its returned boxes agree exactly with the programmatically computed match set. EMSpec-Instruct contains three task families. **Modulation recognition** predicts the modulation category of a single signal. **Structured detection** identifies all signal instances in an observation and returns their bounding boxes and class labels in a predefined format. **Language-conditioned signal grounding** retrieves signals satisfying natural-language constraints, including modulation type, time interval, frequency interval, their combinations, and time-frequency overlap. Positive queries return the boxes of all matched signals, while unmatched conditions produce an empty result. Figure 3 summarizes this construction pipeline together with the resulting task and signal-length distributions.

The final dataset contains 400,000 training instructions and 67,883 test instructions, as in Figure 3. We construct aligned multimodal and waterfall-only configurations for controlled comparison between image-only VLMs and models additionally equipped with the BATok. All partitions are defined at the underlying observation level before instruction generation, ensuring that the same I/Q sequence, its derived waterfall image, and associated paraphrased instructions do not cross the training and test splits. Further construction details, source statistics, task distributions, and validation results are provided in Supplementary Sections S2 and S3.3.

Experiments

Experimental Setup

We evaluate BATok in unified VLM and standalone modulation-recognition settings. For the former, BATok is integrated into Qwen3-VL-2B (Bai et al. 2025) and LLaVA-1.5-7B (Liu et al. 2024). The original VLMs are fine-tuned on image-text instructions, whereas their BATok-enhanced counterparts use signal-text, image-text, and image-signal-text instructions. The main evaluation supplies image-text inputs to the original VLMs and aligned image-signal-text inputs to the enhanced models, using the same observations and train-test partitions. All VLM variants are fine-tuned for five epochs on eight NVIDIA A800 GPUs.

Standalone representation evaluation and ablations use the complete HisarMod2019 (Tekbıyık et al. 2020) and EMSPEC-SIM benchmarks; their aligned subsets within EMSpec-Instruct are denoted HISARMOD-SUB and EMSPEC-SIM-SUB. BATok is followed by a lightweight classification head. All standalone classifiers and task-specific detectors are trained for 100 epochs on one NVIDIA GeForce RTX 4090 under common dataset-specific partitions and preprocessing protocols. Further configurations and expanded results are provided in Supplementary Sections S1 and S3.

Unified Instruction-Conditioned Spectrum Understanding

Table 1 compares each image-based VLM with its BATok-enhanced counterpart to test whether raw I/Q inputs can be incorporated without changing the unified task interface. This comparison reflects the combined effect of BATok integration and mixed-modality training; Supplementary Section S3.2 compares input modalities within the same trained checkpoint. The enhanced models yield the largest gains in modulation recognition: OA increases by 7.01 and 18.56 points for Qwen3-VL-2B and LLaVA-1.5-7B, respectively, with Macro-F1 gains of 5.88 and 16.94 points. Positive-query detection and grounding metrics also improve within both backbones, whereas empty-set accuracy decreases slightly by 0.76 and 0.69 points, revealing a small trade-off in rejecting unmatched queries. Localization outputs remain highly parseable, while the lower modulation-recognition parse rate of Qwen3-VL-2B+BATok confirms that format compliance and recognition accuracy are distinct. The consistent recognition gains across both backbones support BATok as a VLM-agnostic interface, while the smaller localization gains indicate that raw I/Q complements the time-frequency geometry already visible in waterfall images. Supplementary Section S3.4 attributes the Qwen3-VL-2B gain mainly to semantics and the LLaVA-1.5-7B gain to joint geometric-semantic improvements. Section S3.2 also shows that signal-text favors recognition, image-text preserves localization, and joint input provides the best balance across the evaluated tasks.

Dataset-Level Task Feasibility

Task-specific models validate the learnability of the recognition and detection subsets assembled from existing and simulated observations. On HISARMOD-SUB/EMSPEC-SIM-SUB, BATok reaches 49.36% Accuracy and 47.19% Macro-F1; on

Backbone	BATok	Modulation Recognition			Structured Detection			Signal Grounding			
		OA (%) ↑	Macro-F1 (%) ↑	Parse (%) ↑	F1@0.5 (%) ↑	mIoU (%) ↑	Parse (%) ↑	F1@0.5 (%) ↑	mIoU (%) ↑	Parse (%) ↑	Empty Acc. (%) ↑
Qwen3-VL-2B		33.90	35.38	100.00	62.25	94.66	98.08	82.52	94.75	99.99	93.19
Qwen3-VL-2B	✓	40.91	41.26	86.02	64.27	95.20	98.29	83.00	95.24	99.96	92.43
LLaVA-1.5-7B		25.58	26.26	100.00	45.99	88.72	99.43	66.94	88.81	99.95	91.92
LLaVA-1.5-7B	✓	44.14	43.20	94.95	47.80	89.40	97.17	68.27	89.39	99.87	91.23

Table 1: Unified spectrum-understanding results on EMSpec-Instruct. The original VLMs receive image–text inputs, whereas their BATok-enhanced counterparts receive aligned image–signal–text inputs. Parse is the proportion of outputs satisfying the required task-specific format, and Empty Acc. is the accuracy of returning an empty set for unmatched grounding queries. Within each backbone, the better value between the original VLM and its BATok-enhanced counterpart is bolded.

structured detection, YOLO26x reaches 0.9517 mIoU and 0.6925 F1@0.5. These results show that the constructed observations and annotations support their intended tasks. Additional results are provided in Supplementary Section S3.3.

Qualitative Analysis of Detection and Grounding

Figure 4 contrasts the image-only Qwen3-VL-2B baseline with its BATok-enhanced counterpart on representative simulated examples. In the detection example, both models accurately localize the two overlapping transmissions, but the image-only baseline assigns both instances to 16QAM. With access to raw I/Q observations, the BATok-enhanced model recognizes the two ground-truth modulation classes. In the relational grounding example, the image-only baseline selects a different high-energy band, whereas the BATok-enhanced model identifies the longest-duration transmission. These examples illustrate the behaviors reflected by the aggregate metrics rather than serving as substitutes for quantitative evaluation. Additional cases covering both VLM backbones and diverse query types are provided in Supplementary Section S5. Taken together, Figure 4 shows that the raw I/Q branch provides complementary evidence for resolving modulation ambiguity and selecting signals under relational constraints, while the image stream preserves the time–frequency structure required for localization.

Standalone Evaluation of Signal Representations

We next use independently trained signal classifiers to evaluate raw I/Q representation quality without the mixed-modality training procedure of the VLM experiments. This setting directly tests whether BATok learns effective signal representations and how it compares with task-specific signal encoders on complete classification benchmarks. Table 2 compares raw-I/Q signal encoders across two independently trained modulation-recognition benchmarks. The references include MAMC (Zhang et al. 2024), the Signal Transformer (SiT) (Zhai et al. 2025), and DNCNet with its CNN-based signal backbone (Du et al. 2022). These task-specific models are not expected to provide the instruction-following and grounding capabilities of the unified VLM; instead, they provide controlled reference points for representation quality.

BATok obtains the highest mean Accuracy (58.64%) and Macro-F1 (58.54%) across the two benchmarks. It performs

Method	Acc. (%) ↑	Macro-F1 (%) ↑
MAMC (Zhang et al. 2024)	43.20	42.31
SiT (Zhai et al. 2025)	53.56	53.36
DNCNet (Du et al. 2022)	<u>58.18</u>	<u>56.89</u>
BATok	58.64	58.54

Table 2: Standalone raw-I/Q modulation-recognition results. Each average is the unweighted mean across the two primary benchmarks, HisarMod2019 and EMSPEC-SIM. Best and second-best averages are bolded and underlined, respectively. More results are provided in Supplementary Section S4.1.

best on HisarMod2019 and achieves the highest Macro-F1 on EMSPEC-SIM, while DNCNet retains the highest Accuracy on EMSPEC-SIM. Its average gains over the second-best method are 0.46 Accuracy and 1.65 Macro-F1 points. Thus, BATok is not uniformly superior on every dataset, but its bounded representation remains competitive across distinct signal regimes. Per-benchmark and input-configuration results are provided in Supplementary Sections S4.1 and S3.2.

Ablation Studies

To isolate the contribution of individual tokenizers from the language model and multimodal instruction-tuning procedure, we perform all ablations in the standalone modulation-recognition setting. Table 3 evaluates four design questions: how token capacity should depend on signal length, whether signal-derived features improve raw I/Q representations, how many temporal resolutions are needed, and whether the energy prior and learned relevance score provide complementary resampling biases. All variants follow the same evaluation protocol, and the reported metrics are averaged over HisarMod2019 and the complete EMSPEC-SIM benchmark.

Among the selected operating points in Table 3, the piecewise schedule $K(T)$ attains the strongest mean Macro-F1 (58.54%) and remains within 0.06 points of the best mean Accuracy obtained by Linear-high. Signal-derived features improve Accuracy/Macro-F1 by 2.27/2.24 points over I/Q alone, and the third temporal branch adds 2.58/2.46 points over two branches. A similar trend appears in resampling, where combining energy and learned relevance exceeds the

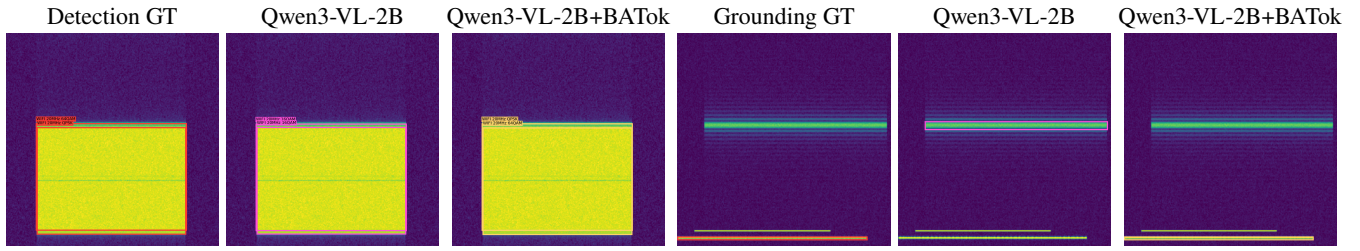


Figure 4: Visualization of detection and grounding results. **Detection question:** “Enumerate every transmission using aligned box and class arrays.” **Ground truth:** [141, 424, 846, 924] (WiFi 20 MHz 64QAM) and [145, 443, 842, 943] (WiFi 20 MHz QPSK). **Grounding question:** “Return boxes of the longest-duration transmissions.” **Ground truth:** [2, 956, 887, 969].

Group	Variant	Acc. (%) $\uparrow$	Macro-F1 (%) $\uparrow$
Token budget	Fixed $K = 32$	57.68	57.53
	Fixed $K = 48$	57.97	57.60
	Linear-high	58.70	<u>58.48</u>
	Piecewise $K(T)^\dagger$	<u>58.64</u>	58.54
Input features	I/Q only	<u>56.37</u>	<u>56.30</u>
	I/Q + $A + P + \Delta\phi^\dagger$	58.64	58.54
Temporal scales	One branch (small)	54.70	54.53
	Two branches (small+mid)	<u>56.06</u>	<u>56.08</u>
	Three branches $\dagger$	58.64	58.54
Resampling bias	Energy prior only	<u>56.97</u>	<u>57.22</u>
	Learned relevance only	56.80	56.45
	Energy + learned relevance $\dagger$	58.64	58.54

Table 3: Ablation study of BATok in standalone modulation recognition. Accuracy and Macro-F1 are percentages averaged over HisarMod2019 and EMSPEC-SIM; $\dagger$ denotes the full BATok configuration within each ablation group.

stronger single bias by 1.67/1.32 points. These results show that length-aware capacity, multi-resolution encoding, and complementary resampling cues jointly support the final representation. Full budget sweeps and per-dataset results are reported in Supplementary Sections S4.1–S4.4.

Conclusion

In this paper, we introduced BATok, which maps mixed-length raw I/Q observations to bounded, content-aware tokens for MLLMs. A length-dependent budget, multi-resolution encoding, and energy–relevance-guided resampling adapt capacity without exposing the language model to unbounded sequences. Across two VLMs, BATok improves modulation recognition and positive-query detection and grounding, showing that waveform evidence complements waterfall images. Standalone results and ablations support its representation and design. However, the limitation is that open-set and cross-domain generalization remain unverified. Moreover, BATok’s small parameter footprint and bounded output coexist with higher standalone latency than compact classifiers because of its multi-resolution resampling.

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Supplementary

S1 Architectural and Implementation Details of BATok

This section specifies the BATok configuration used in the reported experiments. Given a variable-length I/Q sequence, BATok lifts the waveform into signal-derived features, encodes them at multiple temporal resolutions, and produces a bounded token sequence for VLM integration or standalone classification.

S1.1 Input Representation and Normalization

Each signal is represented as $X \in \mathbb{R}^{2 \times T}$, with the first and second channels containing the in-phase and quadrature components. We independently normalize each signal by its mean complex-sample power:

$$\tilde{X} = \frac{X}{\sqrt{\frac{1}{T} \sum_{t=1}^T (I_t^2 + Q_t^2) + 10^{-8}}}. \quad (\text{S1})$$

Signals in a minibatch are zero-padded to the longest signal in that minibatch, while their valid lengths are retained separately. BATok uses these lengths to construct branch masks and to remove padded signal tokens before insertion into the VLM.

S1.2 Feature Lifting and Multi-Resolution Encoding

BATok explicitly lifts the two I/Q channels to five signal-derived channels:

$$U_t = [I_t, Q_t, A_t, P_t, \Delta\phi_t], \quad (\text{S2})$$

where

$$A_t = \sqrt{I_t^2 + Q_t^2 + 10^{-8}}, \quad P_t = I_t^2 + Q_t^2, \quad (\text{S3})$$

and the inter-sample phase difference is implemented without an explicit phase unwrap:

$$\Delta\phi_t = \text{atan2}(Q_t I_{t-1} - I_t Q_{t-1}, I_t I_{t-1} + Q_t Q_{t-1}). \quad (\text{S4})$$

The first phase-difference value is zero-padded. A convolutional stem with kernel size 7, stride 1, Group Normalization (Wu and He 2018), and GELU maps the five channels to a d_s -dimensional temporal representation. In the VLM experiments, $d_s = 256$.

The projected sequence is processed by three parallel temporal branches. Table S1 gives their exact downsampling configurations. Each branch first applies a depthwise-separable convolution (Howard et al. 2017), followed by one LiteMixer block in the VLM experiments. A LiteMixer block alternates a residual depthwise temporal convolution with a residual gated MLP; invalid positions are masked after each block.

Branch	Kernel (samples)	Stride (samples)	Approximate length (samples)
Small	7	4	$\lceil T/4 \rceil$
Mid	31	16	$\lceil T/16 \rceil$
Large	127	64	$\lceil T/64 \rceil$

Table S1: Temporal-branch configuration used by BATok. The output length is approximate because convolution padding and the valid-length mask are handled separately.

Each branch computes its own local-energy sequence directly from the I/Q input rather than from its hidden features:

$$E_b = \text{AvgPool1D}_{k_b, s_b}(I^2 + Q^2). \quad (\text{S5})$$

The pooling kernel k_b and stride s_b match the corresponding temporal branch. Consequently, the small, mid, and large branches have separate energy sequences aligned with their own temporal resolutions.

S1.3 Length-Adaptive Budget and Scale Allocation

BATok determines its total token budget using a monotonic piecewise-constant function of the input length T , as summarized in Table S2.

The three scale budgets are

$$K_s = \lfloor K/4 \rfloor, \quad K_m = \lfloor K/4 \rfloor, \quad K_l = K - K_s - K_m. \quad (\text{S6})$$

Valid input length T (samples)	Total budget $K(T)$ (tokens)
$T \leq 2^{12}$	32
$2^{12} < T \leq 2^{15}$	48
$2^{15} < T \leq 2^{16}$	64
$2^{16} < T \leq 2^{18}$	96
$2^{18} < T \leq 2^{20}$	128
$2^{20} < T \leq 2^{22}$	192
$T > 2^{22}$	256

Table S2: BATok token schedule.

Thus, one quarter of the tokens is assigned to each of the small and mid-branches, and the remaining half is assigned to the large branch. For example, $K = 48$ yields $(K_s, K_m, K_l) = (12, 12, 24)$. The standalone experiments are pure modulation-classification evaluations on the original HisarMod2019 and EMSPEC-SIM benchmarks, from which aligned subsets are used to construct EMSpec-Instruct. The two primary standalone benchmarks likewise activate 32 or 48 tokens; the remaining schedule entries specify BATok’s bounded behavior for longer valid inputs.

After resampling, each branch token receives a learnable scale embedding. The three token sets are concatenated and refined by a shared residual projection consisting of LayerNorm, a d_s -to- d_s linear layer, GELU, and dropout. For mixed-length minibatches, BATok pads this output to the largest token count in the minibatch and returns a Boolean token mask.

S1.4 Placeholder Expansion and Modality Handling

The textual string `<signal>` is used only as an insertion anchor and is not added as a new atomic vocabulary item. During the forward pass, the integration layer locates the tokenizer sequence corresponding to the anchor and replaces its embeddings with the valid projected signal tokens. It simultaneously updates the attention mask, position indices, and labels. Labels at signal-token positions are excluded from the autoregressive objective, so the loss is computed only on the target response.

The integration layer enforces modality consistency: a sample containing a signal anchor must supply an I/Q input, and an I/Q input must have a corresponding anchor. This check prevents a silent shift between the signal batch and the textual batch. It also supports mixed-modality minibatches through per-sample indices, so image–text, signal–text, and image–signal–text examples can be trained together.

S1.5 Backbone-Specific Integration

Table S3 summarizes the current integration settings documented by the training configurations. Both backbones use the same BATok architecture: $d_s = 256$, one LiteMixer block per branch, four resampler heads, zero signal-encoder dropout, a configured query capacity of 512, and the active budget schedule in Table S2. BATok and the signal projector are fully trainable in both models, while the backbone language-model layers are adapted with LoRA (Hu et al. 2022). All VLM variants are fine-tuned for five epochs.

Setting	Qwen3-VL-2B	LLaVA-1.5-7B
LLM adaptation	LoRA ($r = 32$)	LoRA ($r = 8$)
Vision tower	Trainable	Trainable
Native multimodal projector	Trainable	Trainable
BATok	Trainable	Trainable
Signal projector	Trainable	Trainable
Signal position handling	Extended M-RoPE	Learned 1D embedding

Table S3: Backbone-specific VLM integration settings. The language models are adapted with LoRA, while BATok and its projector are trained and saved as complete modules.

For Qwen3-VL (Bai et al. 2025), the integration layer first obtains the native multimodal rotary position indices for image and text tokens. It then inserts consecutive signal positions at the expanded anchor, shifts subsequent positions by the change in sequence length, and updates the multimodal token types. During cached decoding, the attention mask and position indices are reconstructed from the actual prefill-cache length.

For LLaVA-1.5 (Liu et al. 2024), the projected signal tokens receive a learned one-dimensional position embedding before insertion. The described LLaVA experiment initializes BATok from the Qwen3-VL-2B-trained signal encoder, but constructs a new signal projector and signal position embedding because the two backbones need not share a language hidden dimension. The vision tower and native multimodal projector remain trainable.

S1.6 Standalone Classification Head

For standalone signal classification, BATok produces $Z \in \mathbb{R}^{B \times K_{\max}^{(B)} \times d_s}$ and a validity mask M . We aggregate the valid tokens using masked mean pooling:

$$\bar{z}_i = \frac{\sum_k M_{i,k} Z_{i,k}}{\max(1, \sum_k M_{i,k})}. \quad (\text{S7})$$

The pooled representation is passed through LayerNorm, a d_s -dimensional hidden layer, GELU, dropout, and a final classification layer. The training objective is cross-entropy with optional label smoothing. Dataset-specific optimization settings are reported with the corresponding experimental results rather than inferred from architectural defaults. BATok and all task-specific standalone classifiers are trained for 100 epochs.

S2 EMSpec-Instruct Construction Details

This section details the construction procedure summarized in the main paper. EMSpec-Instruct follows an *observation-centered* design: each observation comprises an I/Q sequence, its deterministically rendered waterfall image, and a structured list of signal instances. We assign observations to data partitions before deriving instructions and targets. Consequently, all instructions generated from the same observation remain in one partition, even when a multi-signal scene yields several detection and grounding queries.

S2.1 Source Observations and Partitioning

We use separate single-signal and multi-signal streams. The single-signal stream combines HISARMOD-SUB, a subset of HisarMod2019.1 (hereafter HisarMod2019) (Tekbıyık et al. 2020), with EMSPEC-SIM-SUB, a subset of the complete EMSPEC-SIM single-signal simulation benchmark, and supplies modulation-recognition examples. The multi-signal stream combines SpaceNet (ChaSpark 2026) with our multi-signal simulation and supplies structured-detection and signal-grounding examples. Table S4 summarizes the number of source observations available to the final construction procedure. We preserve the source-provided train/test assignment and perform all subsequent sampling within the corresponding partition. In particular, instructions are never randomly split after generation. The tuple consisting of the source dataset, source split, source sample identifier, and signal path is used as the observation identity. Thus, all instructions derived from one observation remain in the same partition.

Source	Task stream	Category coverage	Train (observations)	Test (observations)
HISARMOD-SUB	Recognition	26 classes: PSK ₆ , QAM ₇ , FSK ₄ , PAM ₃ , and six analog AM/FM/PM variants	44,000	22,000
EMSPEC-SIM-SUB	Recognition	34 classes: generic digital/analog modulations, Wi-Fi ₆ , BLE LE1M/LE2M, LoRa 125/250/500 kHz, Zigbee, and SRRC-shaped QPSK/16QAM	22,000	11,000
SpaceNet	Det./ground.	11 wireless classes: Wi-Fi ₆ , BLE LE1M/LE2M, LoRa 250 kHz, Zigbee, and SRRC-shaped QPSK	7,500	2,500
Multi-signal simulation	Det./ground.	14 classes: the SpaceNet vocabulary augmented with AM, FM, and SRRC-shaped 16QAM	20,000	7,500
Total			93,500	43,000

Table S4: Source observations and category coverage used to construct EMSpec-Instruct. Counts refer to electromagnetic observations before instruction expansion. Wi-Fi₆ denotes the six combinations of 20/40-MHz bandwidth and QPSK/16QAM/64QAM modulation.

Simulation procedure. The simulated observations are generated with RF waveform-generation utilities following the TorchSig simulation framework (Boegner et al. 2022a), from a fixed class map and seeded random state. For the single-signal stream, we sample a class with approximately uniform frequency, synthesize its baseband waveform, apply the class-specific modulation and pulse shaping, and add complex Gaussian noise at an SNR selected from

$$\{-20, -15, -10, -5, 0, 5, 10, 15, 20, 25, 30\} \text{ dB}.$$

Sequence length is sampled from

$$\{1,024, 2,048, 4,096, 8,192, 16,384, 32,768\},$$

while the nominal sample rate is selected from 1, 2, 5, 10, or 20 MHz; wider rates are used when required by Wi-Fi signals. This provides approximately balanced coverage of all 34 simulated classes.

For the multi-signal stream, two or more independently synthesized waveforms are placed in a common time–frequency canvas. We sample the class, active time interval, center frequency, occupied bandwidth, SNR, near–far power offset, and channel–frequency offset for every instance. Placement is stratified over non-overlap, time-only overlap, frequency-only overlap, partial time–frequency overlap, and near-complete overlap configurations. Signals are shifted, gated, scaled, and summed in complex baseband; the clean placement parameters are retained as instance metadata. Bounding boxes are therefore obtained from generation parameters rather than estimated from the rendered image. The simulation complements the source datasets with controlled class balance, signal density, acquisition length, SNR, and overlap difficulty.

S2.2 Instance Metadata

Every signal instance is represented by

$$o_i = (c_i, [t_i^s, t_i^e], [f_i^l, f_i^h], b_i, a_i), \quad (\text{S8})$$

where c_i is the canonical class name, the next two terms are its time and frequency intervals, b_i is its normalized bounding box, and a_i contains optional attributes such as SNR, signal family, modulation, or communication standard. Class aliases are canonicalized before sampling; for example, order-reversed names such as QAM16 and 16QAM map to the same category.

S2.3 Time–Frequency Box Conversion

Let an observation span $[0, D]$ milliseconds and $[F_{\min}, F_{\max}]$ MHz. For an instance with time interval $[t_i^s, t_i^e]$ and frequency interval $[f_i^l, f_i^h]$, its continuous normalized coordinates are

$$\begin{aligned} x_1 &= 1000 t_i^s / D, & x_2 &= 1000 t_i^e / D, \\ y_1 &= 1000 \left(1 - \frac{f_i^h - F_{\min}}{F_{\max} - F_{\min}} \right), & y_2 &= 1000 \left(1 - \frac{f_i^l - F_{\min}}{F_{\max} - F_{\min}} \right). \end{aligned} \quad (\text{S9})$$

The inversion of the frequency coordinate is consistent with the image convention that high frequency appears at the top. Coordinates are rounded to integers, clamped to $[0, 1000]$, and sorted along each axis. We reject boxes with non-positive width or height. The coordinate representation is independent of the raster resolution, which allows images to be resized by the VLM processor without changing the targets.

All time- or frequency-conditioned questions explicitly state the physical observation range. For image-containing examples, the standard context describes time from left to right and frequency from bottom to top. It also defines the box ordering as $[x_{\min}, y_{\min}, x_{\max}, y_{\max}]$. Signal-only prompts contain the same numerical ranges without referring to visual axes. This context prevents a model from having to infer milliseconds or MHz from normalized box coordinates alone.

S2.4 Task Construction

We first construct a deterministic semantic specification for every instruction. This specification contains the task family, query subtype, condition, matched instance identifiers, exact target boxes, and negative-query flag. Natural-language realization is performed only after this specification has been fixed.

Modulation recognition. Each single-signal observation supplies a class-prediction semantic specification. Seeded quota sampling and modality packaging may produce multiple instruction records from the same observation, but all such records remain in the same data partition. The training subset is class-balanced over 50 modulation and communication-signal categories, with 2,400 instructions per class and 120,000 instructions in total. Qwen3.5-Max (Team 2026) generates diverse recognition questions from the task name and modality configuration; the answer is then filled directly with the canonical class label from the source annotation. The target is a one-element class list.

Structured detection. Each multi-signal observation yields one all-signal detection instruction. The target contains every valid instance in the scene. Boxes and class labels are sorted in the same instance order so that the j -th box corresponds to the j -th class. The multi-signal class universe contains 14 classes: AM, FM, BLE LE1M, BLE LE2M, LoRa 250KHZ, SRRC QPSK, SRRC 16QAM, six WiFi bandwidth/modulation combinations, and Zigbee. As in recognition, Qwen3.5-Max generates the instruction, whereas the complete ordered box/class response is serialized directly from the scene annotation.

Language-conditioned signal grounding. Grounding instructions return the boxes of all instances satisfying a language condition. We use the following condition families.

- *Class*: exact match after class-name canonicalization.
- *Time or frequency*: non-empty interval intersection between the instance and the requested range.
- *Class–time–frequency composition*: conjunction of the class and both interval conditions.
- *Duration and bandwidth relations*: all instances tied for the longest duration or widest occupied bandwidth.
- *Waterfall strength*: strongest or weakest instances according to image evidence.
- *Overlap*: instances whose time and frequency intervals both intersect another instance.

The waterfall-strength attribute is computed rather than obtained from a language model. Specifically, we use the 90th percentile of grayscale intensity inside each normalized instance box. Using a high percentile is more robust than a maximum, which can be dominated by a single bright pixel, and less likely than the mean to be diluted by background pixels within a loose box.

Negative grounding. Negative examples are part of signal grounding rather than a separate task. We construct an absent-class query by selecting a class outside the scene’s class set. We also construct a class-conditioned time–frequency query whose conjunction matches no instance. Both types have a deterministic empty target. This forces the model to distinguish failure to satisfy a condition from failure to follow the requested output format.

The training policy first creates six positive and two negative grounding instructions per multi-signal scene, then adds positive subtypes to meet the target task distribution. The held-out localization set contains one or two positive questions and one absent-class question per scene. Its positive subset additionally contains overlap questions when overlapping instances exist; scenes without an overlap may contribute a class-conditioned fallback question.

S2.5 Metadata-Conditioned Instruction Realization

All natural-language instruction records are realized with Qwen3.5-Max. Its role depends on the task. For modulation recognition and structured detection, the model generates only the question; the compact answer is serialized directly from the canonical class label or complete scene annotation. For signal grounding, it generates both the metadata-conditioned question and answer. A grounding record is retained only when its returned boxes and classes agree exactly with the programmatically computed match set. Qwen3.5-Max therefore controls linguistic realization but cannot alter the supervision.

We invoke Qwen3.5-Max through a two-message chat interface. The `system` message assigns an RF-domain role and global faithfulness constraints. The `user` message contains the task-specific instruction, scene duration, observed frequency range, and a compact list of instances. Every instance includes its canonical class, normalized box, time/frequency intervals, and optional SNR or protocol attributes. The user message also contains the requested condition, expected match set, and output schema. Accordingly, the complete input has four logical parts: system role, task-specific instruction, scene metadata, and output constraints. The system message is provided verbatim below:

You are an expert in radio-frequency signal analysis, electromagnetic spectrum monitoring, and time-frequency signal grounding. Your task is to generate accurate and diverse English query-answer pairs from the provided signal metadata. Strictly follow the supplied class labels, time ranges, frequency ranges, SNR values, and normalized bounding boxes. Do not infer or invent any signal, attribute, coordinate, or physical range that is not present in the metadata. Return valid JSON only.

For this input, a valid generated query is: “Within the 150-ms observation spanning 2430–2480 MHz, locate signals intersecting 2470–2474 MHz.” The generated answer mentions both matched classes and contains exactly `<box>93, 157, 360, 163</box>` and `<box>547, 60, 1000, 860</box>`. A recognition input is shorter: it provides the task, available modalities, and canonical label, and asks Qwen3.5-Max for a concise modulation-classification question. A detection input provides the scene context and all instance metadata and asks only for a diverse formulation of “detect all signals”; its answer is not generated by Qwen3.5-Max.

The Qwen3.5-Max output is accepted only when all of the following checks pass: (1) the number and order of returned items match the requested items; (2) every query is non-empty and retains explicit millisecond/MHz scene context when required; (3) class names and numerical condition ranges are consistent with the semantic specification; (4) grounding answers contain every and only expected box, while negative answers contain no box and explicitly state that no matching signal exists; and (5) the response parses as the requested JSON object. The model is forbidden from inventing classes, boxes, physical ranges, or attributes. Only validated Qwen3.5-Max records enter the dataset.

S2.6 Compact Target Grammar

All targets use a compact, machine-parseable grammar. Let $B = [[x_1, y_1, x_2, y_2], \dots]$ be a list of boxes and $C = [c_1, \dots]$ the corresponding class list. The three task families use

$$\begin{aligned} \text{modulation recognition:} & \quad \text{<c>C</c>}, \\ \text{structured detection:} & \quad \text{<d>B<c>C</c></d>}, \\ \text{signal grounding:} & \quad \text{B}. \end{aligned} \tag{S10}$$

For a negative grounding query, B is the empty list and the exact target is `<b>[]</b>`. This grammar separates semantic evaluation from free-form wording: modulation classes, boxes, and box–class correspondence can be parsed without another language model. Table S5 gives representative examples with numerical values shortened for readability.

S2.7 Sampling, Modalities, and Final Statistics

The final training set contains 400,000 instruction records. We allocate 30% to single-signal modulation recognition and 70% to multi-signal understanding. The latter is divided into one detection instruction per source scene and multiple grounding instructions. Table S6 reports the exact train and test sizes.

Table S7 gives the grounding subtype distribution. The training counts reflect quota-based sampling rather than the raw number of candidate questions. The “class fallback” test subtype is used only when an overlap query cannot be constructed.

Task	Example instruction	Target
Recognition	Classify the modulation using both modalities.	<c> ["8ASK"] </c>
Detection	Detect every signal and return all boxes and classes.	<d> [[339, 166, 982, 833]] <c> ["16QAM"] </c></d>
Grounding	Find regions that overlap 2419.589–2448.466 MHz.	 [[339, 166, 982, 833], . . .]
Negative grounding	Find all BLE LE2M regions in this scene.	 []

Table S5: Representative task realizations and target formats. Scene context is abbreviated in the table but is present in the actual time/frequency-conditioned prompts.

Task	Train (records)	Train (%)	Test (records)	Test (%)
Modulation recognition	120,000	30.00	33,000	48.61
Structured detection	27,500	6.88	10,000	14.73
Signal grounding	252,500	63.12	24,883	36.66
Total	400,000	100.00	67,883	100.00

Table S6: Final task composition. The validation set is a balanced 3,000-example subset sampled from the held-out task files and is not included in the test total.

Grounding subtype	Train (records)	Test (records)
Class	27,500	1,631
Class fallback (no overlap)	–	313
Time	27,500	1,656
Frequency	27,500	1,640
Class–time–frequency	27,500	1,656
Longest duration	27,500	1,638
Widest bandwidth	27,500	1,689
Strongest brightness	27,500	1,617
Weakest brightness	5,000	1,692
Time–frequency overlap	–	1,351
Negative class	27,500	10,000
Negative class–condition	27,500	–
Total	252,500	24,883

Table S7: Signal-grounding subtype counts. A dash denotes a subtype that is not selected for that partition.

We construct aligned *image* and *signal* JSONL configurations. The image configuration presents every record as waterfall-image plus text. The signal configuration is used to train the unified model and contains 298,000 joint image–signal records, 74,000 signal-only records, and 28,000 image-only records. The underlying task target, source sample identifier, and train/test membership are identical across configurations. This alignment supports controlled comparison between an image-only VLM and a VLM equipped with BATok. Including image–text, signal–text, and image–signal–text instructions in one training set exposes the model to the same task interface under different modality configurations.

The data cover a wide range of observation lengths. After deduplicating repeated instructions by source observation, the training partition contains 93,500 signal observations with lengths from 1,024 to 15,000,000 complex samples. Its median is 1,024, the 95th percentile is 6,000,000, and the mean is approximately 1.05 million samples.

S3 Extended Experimental Evaluation

S3.1 Evaluation Metrics

For modulation recognition, we report overall accuracy (OA) and Macro-F1. OA measures the proportion of correctly classified samples, whereas Macro-F1 averages class-wise F1 scores and therefore reflects performance across modulation classes without allowing frequent classes to dominate the result. For VLM outputs, we additionally report the parse rate, defined as the proportion of responses that conform to the required task-specific output format. This separates semantic task performance from the ability to follow the structured response protocol.

For structured detection and positive-query grounding, we use F1@0.5 and mean intersection over union (mIoU). A predicted box is considered matched when its IoU with a ground-truth box is at least 0.5. Detection matching is class-aware and

Backbone	Test Input	Mod. Rec.			Detection			Grounding			
		OA (%) ↑	Macro-F1 (%) ↑	Parse (%) ↑	F1@0.5 (%) ↑	mIoU (%) ↑	Parse (%) ↑	F1@0.5 (%) ↑	mIoU (%) ↑	Parse (%) ↑	Empty Acc. (%) ↑
Qwen3-VL-2B + BATok	Image + Text	16.52	11.93	100.00	64.12	95.20	97.88	81.78	95.23	99.94	92.04
Qwen3-VL-2B + BATok	Signal + Text	40.36	40.33	88.96	3.64	72.48	99.44	13.36	74.37	99.90	89.39
Qwen3-VL-2B + BATok	Image + Signal + Text	40.91	41.26	86.02	64.27	95.20	98.29	83.00	95.24	99.96	92.43
LLaVA-1.5-7B + BATok	Image + Text	3.08	2.54	99.35	47.69	89.28	98.19	67.75	89.29	99.88	85.15
LLaVA-1.5-7B + BATok	Signal + Text	43.68	43.02	94.90	3.06	74.68	98.80	12.85	75.05	99.81	89.11
LLaVA-1.5-7B + BATok	Image + Signal + Text	44.14	43.20	94.95	47.80	89.40	97.17	68.27	89.39	99.87	91.23

Table S8: BATok-enhanced VLM performance by input configuration. Values are percentages, and all input configurations for a backbone use the same trained checkpoint. Parse denotes successful task-format parsing, and Empty Acc. is reported for unmatched grounding queries.

additionally requires the correct signal label, whereas grounding evaluates the boxes selected by the query and does not involve a class prediction. F1@0.5 jointly captures missed and spurious targets through precision and recall, while mIoU measures the spatial agreement of matched boxes. We report precision and recall separately for the task-specific detectors to expose their false-positive and false-negative trade-off. For unmatched grounding queries, where no target box exists, empty-set accuracy measures whether the model correctly returns no box.

The gain-decomposition analysis further reports class-agnostic F1@0.5 and matched-box mIoU together with conditional class accuracy, allowing target discovery and box geometry to be separated from signal-category prediction. Class-agnostic mAP@0.5 and mAP@0.5:0.95 provide complementary precision-recall summaries across localization thresholds; because the generated boxes do not carry confidence scores, their output order is used for this secondary analysis. For efficiency comparisons, parameter count measures model size, latency measures batch-size-one inference time in milliseconds per sample, and the realized token count K measures the representation budget presented to the downstream model.

S3.2 Spectrum Understanding Across Input Configurations

Table S8 isolates the contribution of each input pathway. Each BATok-enhanced VLM is trained as one checkpoint with image-text, signal-text, and image-signal-text instructions. At test time, the three input configurations share the same held-out observations, prompts, and targets, so their differences directly reflect the evidence supplied by each modality.

The same-checkpoint comparison separates the contribution of test-time signal input from the overall effect of BATok integration and mixed-modality training. For Qwen3-VL-2B+BATok, image-text input already achieves 64.12% detection F1, and adding signal input raises it to 64.27%. The increase from 62.25% to 64.27% in Table 1 therefore reflects the overall change in the model and training setup, whereas the increase from 64.12% to 64.27% measures the additional contribution of waveform evidence within the enhanced checkpoint.

The modalities play different roles across tasks. Signal-text input achieves recognition OA of 40.36% and 43.68% for Qwen3-VL-2B and LLaVA-1.5-7B, respectively, while its detection and grounding F1 remain low. Image-text input provides the main spatial evidence for localization. Joint input raises recognition OA to 40.91% and 44.14% and improves grounding F1 over image-text input by 1.22 and 0.52 percentage points, respectively. These results support complementary roles for waveform evidence and waterfall geometry within the unified instruction interface.

S3.3 Validation of EMSpec-Instruct

We evaluate whether the recognition and detection subsets of EMSpec-Instruct provide valid, learnable supervision for their intended tasks. Task-specific classifiers and detectors assess the observations, class labels, and bounding-box annotations assembled from existing and simulated signal sources, while VLM+BATok evaluates whether the same data can support a unified instruction-conditioned interface. The aligned recognition experiment additionally reveals how the raw I/Q observations and their derived waterfall images preserve modulation-class evidence.

Multi-Task Instruction Supervision We evaluate whether the three task families in EMSpec-Instruct provide complementary supervision for language-conditioned signal grounding. The Qwen3-VL-2B+BATok architecture and grounding test set are fixed, while the instruction-tuning data contain grounding alone or grounding augmented with modulation recognition, structured detection, or both. Table S9 measures how these complementary task annotations improve grounding.

The complete three-task mixture achieves the highest grounding F1@0.5 (83.00%) and ties the highest mIoU (95.24%). Compared with grounding-only training, jointly incorporating modulation-recognition and structured-detection supervision improves F1@0.5 by 1.03 percentage points and mIoU by 0.09 percentage points. These results indicate that joint training on the three tasks helps improve language-conditioned signal grounding in this experimental setting.

Training tasks			Grounding	
Mod. Rec.	Detection	Grounding	F1 @0.5 (%) ↑	mIoU (%) ↑
		✓	81.97	95.15
✓		✓	81.99	95.24
	✓	✓	81.47	95.06
✓	✓	✓	83.00	95.24

Table S9: Grounding performance under different instruction-tuning task compositions. All variants use Qwen3-VL-2B+BATok and are evaluated on the same grounding test set.

Modulation-Recognition Learnability Table S10 evaluates task-specific modulation recognition on the union of HISARMOD-SUB and EMSPEC-SIM-SUB, the two aligned modulation-recognition subsets of EMSpec-Instruct. Its primary purpose, together with the structured-detection evaluation below, is to verify that the constructed observations and annotations support learnable recognition and detection tasks. We include classifiers operating on both waterfall images and raw I/Q observations. Because the two modalities are derived from the same source observations and use the same partitions, their results also indicate how much modulation-class evidence is retained by each representation. The image references use ResNet-50 (He et al. 2016) and ViT (Dosovitskiy et al. 2021); the raw-I/Q references include MAMC (Zhang et al. 2024), the Signal Transformer (SiT) (Zhai et al. 2025), and DNCNet with its ResNet18-1D signal backbone (Du et al. 2022).

Method	Input	Acc. (%) ↑	Macro-F1 (%) ↑	Params (M) ↓	Latency (ms/sample) ↓
ResNet-50	Waterfall	32.43	34.33	23.610	4.105
ViT	Waterfall	31.06	33.13	85.837	3.424
MAMC (Zhang et al. 2024)	I/Q	29.62	27.64	0.009	1.466
SiT (Zhai et al. 2025)	I/Q	41.56	41.43	0.654	0.967
DNCNet (Du et al. 2022)	I/Q	44.17	42.80	3.870	1.860
BATok	I/Q	49.36	47.19	1.047	5.443

Table S10: Waterfall-image and raw-I/Q modulation recognition on the aligned EMSpec-Instruct subset, defined as the union of HISARMOD-SUB and EMSPEC-SIM-SUB. Latency is measured in milliseconds per sample with batch size 1 on one NVIDIA GeForce RTX 4090.

The image- and raw-I/Q-based classifiers both recover modulation structure from the aligned subsets, confirming that the constructed observations and class annotations support task-specific recognition. BATok records 49.36% accuracy and 47.19% Macro-F1, whereas the strongest waterfall-image result reaches 32.43% accuracy and 34.33% Macro-F1. Beyond validating the dataset, this consistent advantage of the raw-I/Q classifiers supports the observation that the source signals retain modulation-class evidence that is not fully preserved in their derived waterfall images. BATok obtains this result with 1.047M parameters, fewer than DNCNet and far fewer than the image encoders, while its 5.443-ms latency is higher. The difference arises because parameter count measures stored learnable weights, whereas latency is dominated by operations over the input sequence: BATok applies compact shared weights across multiple temporal resolutions and then performs evidence-aware resampling over the resulting features. These length-dependent operations and their memory access make BATok small in parameter count but more compute-intensive at the tokenizer stage. The dedicated BATok classifier also exceeds Qwen3-VL-2B+BATok and LLaVA-1.5-7B+BATok by 8.45 and 5.22 Accuracy points, respectively, providing a direct measure of the signal representation before it is used by the unified generative interface.

Structured-Detection Learnability We train an Ultralytics YOLO implementation (Sapkota and Karkee 2026) and DETR (Carion et al. 2020) for 100 epochs on the structured detection subset of EMSpec-Instruct. Their dedicated detection heads provide task-specific references for evaluating the consistency and learnability of the constructed boxes and class annotations.

Method	mIoU (ratio) ↑	F1 @0.5 (ratio) ↑	Precision@0.5 (ratio) ↑	Recall@0.5 (ratio) ↑
YOLO26n	0.9291	0.6466	0.6288	0.6654
YOLO26l	<u>0.9475</u>	<u>0.6865</u>	<u>0.6514</u>	<u>0.7255</u>
YOLO26x	0.9517	0.6925	0.6577	0.7311
DETR	0.8597	0.4127	0.2918	0.7044

Table S11: Task-specific detection results on the EMSpec-Instruct detection test set. Best and second-best values are bolded and underlined, respectively. These systems use dedicated detection heads.

YOLO26x provides the strongest task-specific detection result, reaching an mIoU of 0.9517 and F1@0.5 of 0.6925. DETR obtains 0.8597 and 0.4127, respectively, while its higher recall than precision indicates more false-positive predictions. The consistent performance across YOLO variants and DETR confirms that the constructed boxes and class annotations provide learnable supervision for structured detection. Detection through the unified generative interface is evaluated separately in Table S8.

S3.4 Analysis of BATok Performance Gains

We further separate modulation semantics from box geometry in the structured-detection results. The main metric is class-aware: a prediction is a true positive only when its class is correct and its IoU is at least 0.5. We additionally recompute class-agnostic metrics by matching boxes solely by IoU, and report conditional class accuracy over predictions already matched to a ground-truth box at $\text{IoU} \geq 0.5$. Table S12 applies all three views to the same predictions used in the main comparison.

Backbone	Detection measure	Image + Text (%)	Image + Signal + Text (%)
Qwen3-VL-2B	Class-aware F1@0.5	62.25	64.27
	Class-agnostic F1@0.5	90.74	90.68
	Conditional class accuracy	68.57	70.72
	Class-agnostic matched-box mIoU	95.02	95.48
	Class-agnostic mAP@0.5	82.87	82.66
	Class-agnostic mAP@0.5:0.95	71.23	72.19
LLaVA-1.5-7B	Class-aware F1@0.5	45.99	47.80
	Class-agnostic F1@0.5	72.58	75.30
	Conditional class accuracy	61.43	62.38
	Class-agnostic matched-box mIoU	88.48	89.20
	Class-agnostic mAP@0.5	54.89	58.60
	Class-agnostic mAP@0.5:0.95	37.02	40.68

Table S12: Decomposition of structured-detection gains into class semantics and box geometry. Class-agnostic metrics ignore the predicted class during IoU matching; conditional class accuracy is computed only over boxes matched at $\text{IoU} \geq 0.5$.

For Qwen3-VL-2B, adding BATok raises class-aware F1 by 2.02 points while class-agnostic F1 remains essentially unchanged. Conditional class accuracy instead increases by 2.15 points, and matched-box mIoU rises by only 0.46 points. Its detection gain therefore comes primarily from more accurate signal-category predictions, with a smaller improvement in box tightness. For LLaVA-1.5-7B, class-agnostic F1 and mAP@0.5 increase by 2.72 and 3.71 points, whereas conditional class accuracy improves by 0.95 points. Here, BATok mainly improves target discovery and localization while also providing a smaller semantic benefit. The source of the gain is thus backbone dependent: predominantly semantic for Qwen3-VL-2B and jointly geometric–semantic for LLaVA-1.5-7B.

Grounding contains no class prediction and is therefore outside this class–geometry decomposition. In the main results, BATok produces smaller, non-uniform grounding changes: matched-box mIoU increases by 0.49 and 0.58 points for Qwen3-VL-2B and LLaVA-1.5-7B, while empty-set accuracy decreases by 0.76 and 0.69 points. BATok should therefore be interpreted as complementary signal evidence rather than a uniform improvement to every localization behavior. Because the generated boxes have no confidence scores, the mAP calculation uses output order; we consequently interpret it jointly with F1, conditional class accuracy, and matched-box mIoU.

S3.5 Fine-Grained Grounding Analysis

Empty-set accuracy is reported only for unmatched queries, for which no ground-truth box exists; it is not substituted for an IoU-based localization metric. Table S13 reports this breakdown for both BATok-enhanced VLM backbones.

Qwen3-VL-2B+BATok exceeds its LLaVA counterpart in F1 across every positive grounding subtype, with the largest F1 among its own subtypes on time-conditioned queries (90.28%). Both models maintain near-perfect parse rates, showing reliable instruction following across class-, time-, frequency-, compositional, relational, and negative queries. The variation across query types is therefore driven primarily by query-conditioned localization.

S4 Extended Ablation and Efficiency Analysis

S4.1 Benchmark-Level Classification Performance and Efficiency

Table S14 expands the averaged results in the main paper into HisarMod2019 and EMSPEC-SIM results, together with parameter count and measured runtime. All models are independently trained for modulation classification on the complete source benchmarks; the aligned source subsets used for dataset validation are evaluated separately in Table S10.

Query Type	Qwen3-VL-2B + BATok				LLaVA-1.5-7B + BATok			
	F1@0.5 (%) ↑	mIoU (%) ↑	Empty Acc. (%) ↑	Parse (%) ↑	F1@0.5 (%) ↑	mIoU (%) ↑	Empty Acc. (%) ↑	Parse (%) ↑
Modulation-conditioned	83.51	94.85	–	99.94	66.59	87.95	–	100.00
Time-conditioned	90.28	95.22	–	99.88	74.07	89.24	–	99.58
Frequency-conditioned	87.02	94.82	–	100.00	74.16	88.91	–	99.88
Compositional	89.46	94.50	–	99.94	79.49	88.80	–	99.88
Overlap / relational	78.00	95.58	–	99.92	66.13	90.01	–	99.87
Unmatched / negative	–	–	92.43	100.00	–	–	91.23	99.88

Table S13: Language-conditioned grounding results by query type for the image–signal–text test configuration. Values are percentages; a dash marks an inapplicable metric, and Parse denotes successful required-format parsing. Overlap/relational pools duration, bandwidth, brightness, and overlap queries, while unmatched pools negative-class and class-fallback-no-overlap queries.

Method	HisarMod2019		EMSPEC-SIM		Params (M) ↓	Latency (ms/sample) ↓
	Acc. (%) ↑	Macro-F1 (%) ↑	Acc. (%) ↑	Macro-F1 (%) ↑		
MAMC (Zhang et al. 2024)	55.22	53.23	31.18	31.39	0.008	1.470
SiT (Zhai et al. 2025)	71.34	71.30	35.78	35.41	0.649	0.978
DNCNet (Du et al. 2022)	<u>75.25</u>	<u>75.19</u>	41.11	<u>38.58</u>	3.856	1.868
BATok	77.53	77.53	<u>39.75</u>	39.54	1.044	5.509

Table S14: Per-benchmark standalone raw-I/Q modulation-recognition results and classification efficiency. Best and second-best performance values within each benchmark are bolded and underlined, respectively. Latency is measured with batch size 1 on one NVIDIA GeForce RTX 4090.

BATok performs best on HisarMod2019 and obtains the highest Macro-F1 on EMSPEC-SIM, while DNCNet obtains the highest Accuracy on EMSPEC-SIM. With 1.044M parameters, BATok is substantially smaller than DNCNet while achieving stronger Macro-F1 on both benchmarks. Its 5.509-ms standalone latency is higher because runtime depends on sequence-level computation rather than parameter count alone. BATok repeatedly applies compact shared operators along long I/Q sequences at multiple resolutions and performs content-aware resampling; these operations increase computation and memory movement without introducing many learnable weights. BATok is therefore small in parameter count and bounded in output length, while allocating additional tokenizer computation to preserve multi-scale signal evidence.

The parameter and latency values differ slightly from those in the aligned-subset validation because the two experiments use different class heads and input-length distributions. Table S10 evaluates the aligned HISARMOD-SUB $\cup$ EMSPEC-SIM-SUB union, whereas Table S14 evaluates the complete standalone benchmarks.

S4.2 Comparison of Token-Budget Strategies

We compare five fixed budgets, three length-dependent linear maps, and the piecewise schedule used by BATok. Table S15 reports average and maximum token counts, cross-dataset mean Accuracy/Macro-F1, and measured latency. The three linear baselines represent increasingly aggressive growth in token capacity with signal length.

Strategy	Avg. K (tokens)	Max K (tokens)	Avg. Acc. (%)	Avg. F1 (%)	Latency (ms/sample)
Fixed-16	16.00	16	57.91	57.62	5.358
Fixed-32	32.00	32	57.68	57.53	5.294
Fixed-48	48.00	48	57.97	57.60	5.300
Fixed-64	64.00	64	58.04	57.70	5.310
Fixed-128	128.00	128	58.08	57.90	5.469
Linear-low	32.50	48	58.60	58.34	5.469
Linear-mid	48.50	64	59.05	58.67	5.536
Linear-high	81.50	128	<u>58.70</u>	58.48	5.456
Piecewise $K(T)^\dagger$	40.00	48	58.64	<u>58.54</u>	5.509

Table S15: Unified performance–budget–latency ablation. Mean Accuracy and Macro-F1 are the unweighted means across HisarMod2019 and EMSPEC-SIM. Latency is measured in milliseconds per sample on one NVIDIA GeForce RTX 4090. $\dagger$ denotes the full BATok configuration. Best and second-best mean results are bolded and underlined, respectively.

The budget sweep shows that increasing the token count alone does not consistently improve classification performance. Across the fixed-budget settings, mean Macro-F1 ranges from 57.53% to 57.90%. Linear-mid achieves the highest mean Accuracy and Macro-F1, at 59.05% and 58.67%, respectively. The piecewise schedule achieves 58.64% Accuracy and 58.54% Macro-F1 with 40 tokens on average, using 17.5% fewer tokens than Linear-mid at decreases of 0.41 and 0.13 percentage points. Compared with Fixed-48, it reduces the average token count by 16.7% while improving Accuracy and Macro-F1 by 0.67 and 0.94 percentage points. These results support the piecewise schedule as a compact configuration that balances token budget and classification performance.

S4.3 Length-Stratified Comparison under Matched Token Budgets

We further isolate signal length on EMSPEC-SIM-VARLEN, a variable-length subset of EMSPEC-SIM. Although samples are stored with zero-padding to 32,768 points, their natural valid lengths are $T \in \{1,024, 2,048, 4,096, 8,192, 16,384, 32,768\}$. All strategies use the same stratified training/validation split and the same untouched test split. Equal-length samples are grouped only to accelerate minibatch processing; no sample is cropped or moved between splits. The piecewise schedule assigns 32 tokens to the first three length groups and 48 tokens to the remaining groups, yielding a sample-weighted test-set average of 39.23 tokens. Fixed $K = 39$ is therefore the matched-budget reference.

Strategy	Avg. K (tokens)	$T = 1,024$	$T = 2,048$	$T = 4,096$	$T = 8,192$	$T = 16,384$	$T = 32,768$
		(samples)	(samples)	(samples)	(samples)	(samples)	(samples)
Fixed-32	32.00	31.99 /30.56	34.69/33.34	38.72 /36.74	41.28/39.43	40.39 /38.67	40.76/39.24
Fixed-39	39.00	31.73/ 30.77	36.19/ 35.39	38.10/ 37.21	41.40/39.97	38.93/37.96	41.18/ 40.70
Fixed-48	48.00	29.23/28.76	34.27/34.04	35.05/34.77	40.62/ 40.35	39.54/ 39.25	39.86/39.23
Piecewise $K(T)^\dagger$	39.23	31.84/29.70	36.73 /34.54	37.35/35.44	42.11 /38.93	40.39 /37.97	41.42 /38.99

Table S16: Controlled within-dataset token-budget comparison. EMSPEC-SIM-VARLEN is a variable-length subset of EMSPEC-SIM. Each length column reports Accuracy/Macro-F1 in percent; Avg. K is weighted by the number of test samples, and bold marks the best result for each metric at each length.

At nearly the same average budget, the piecewise schedule attains 37.67% overall Accuracy versus 37.31% for Fixed-39. Its Accuracy is higher in all three long-input groups, by 0.71, 1.46, and 0.24 points at $T = 8,192, 16,384$, and $32,768$, respectively. Fixed-39 obtains higher overall Macro-F1 (36.46% versus 35.46%), while the piecewise schedule consistently improves Accuracy for long observations. Together, the matched-budget results confirm the intended effect of length-dependent allocation: it directs additional capacity to long signals without increasing the average token budget.

S4.4 Component-Level Ablation Analysis

Table S17 reports per-dataset classification results for input features, temporal scales, and resampling bias. The temporal-scale rows compare the strongest single- and two-branch configurations with the full three-branch model. In each group, $\dagger$ marks the full BATok configuration.

Group	Variant	HisarMod2019 Acc./F1 (%)	EMSPeC-SIM Acc./F1 (%)	Avg. Acc. (%)	Avg. F1 (%)
Input features	I/Q only	74.07/73.97	38.66/38.62	56.37	56.30
	I/Q+A+P+ $\Delta\phi$ †	77.53/77.53	39.75/39.54	58.64	58.54
Temporal scales	Small	68.89/68.88	40.51/40.18	54.70	54.53
	Small+mid	72.64/72.53	39.48/39.63	56.06	56.08
	Small+mid+large †	77.53/77.53	39.75/39.54	58.64	58.54
Resampling bias	Energy prior only	76.99/77.07	36.95/37.36	56.97	57.22
	Learned relevance only	76.88/76.83	36.71/36.06	56.80	56.45
	Energy+learned relevance †	77.53/77.53	39.75/39.54	58.64	58.54

Table S17: Per-dataset component ablations for standalone BATok classification. Dataset columns report Accuracy/Macro-F1 in percent; Avg. columns are arithmetic means across HisarMod2019 and EMSPeC-SIM. † denotes the full BATok configuration, and bold marks the best cross-dataset mean within each ablation group.

The three resampling-bias variants retain the same classifier architecture and each has 1.044M parameters; the ablation changes the evidence score used during resampling.

The full configuration achieves the strongest two-benchmark mean in all three ablation groups. Signal-derived amplitude, power, and phase difference features improve mean Accuracy/Macro-F1 by 2.27/2.24 points over I/Q alone; the third temporal branch adds 2.58/2.46 points over two branches; and combining energy with learned relevance improves by 1.67/1.32 points over the stronger single-bias variant. These consistent gains validate enhanced signal features, multi-resolution encoding, and complementary resampling cues as the three key components of BATok.

S5 Visualization Analysis

The main paper provides one detection case and one grounding case for the Qwen3-VL-2B backbone. Here, we expand the comparison to both Qwen3-VL-2B and LLaVA-1.5-7B and cover narrow-band targets, crowded multi-class scenes, and class-, frequency-, time-, and overlap-conditioned queries. We also retain a case in which all models succeed (Figure S4) instead of selecting only examples favorable to BATok. These examples are intended to expose representative output behavior; performance claims are based on the full-test-set metrics reported in the main paper. The visualizations are read along two complementary axes: whether the image pathway preserves the scene geometry needed for localization, and whether the raw-I/Q pathway resolves signal semantics that remain ambiguous in a waterfall image. Accordingly, each group shows the ground truth and paired predictions from the same prompt and observation, with and without BATok.

S5.1 Structured Detection

Figures S1 and S2 show two complementary challenges. The first combines a broadband WiFi transmission with an extremely narrow LoRa signal, while the second contains four transmissions from three signal families with substantial time–frequency overlap. Each model is evaluated on the same observation and prompt as its paired counterpart.

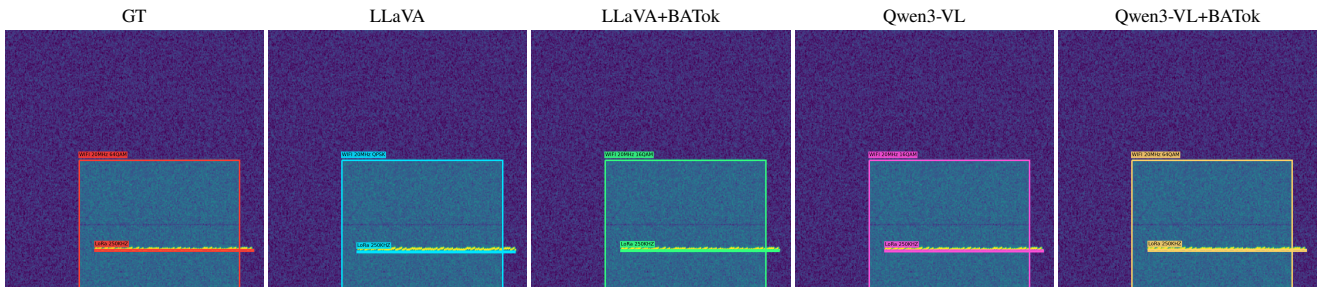


Figure S1: Detection of a broadband transmission and a narrow-band target. **Question:** “Detect all occupied signal regions and pair each bounding box with the matching class label.” **Ground truth:** [285, 500, 904, 1000] (WiFi 20 MHz 64QAM) and [346, 843, 958, 849] (LoRa 250 kHz). Coordinates are normalized to [0, 1000]; the GT panel uses red boxes.

Across both scenes, the image pathway generally recovers the time–frequency extent of broadband, narrow-band, and overlapping transmissions, while the remaining errors are concentrated in their class assignments. Adding BATok preserves these localized regions and provides additional evidence for distinguishing modulation variants. This effect is particularly clear

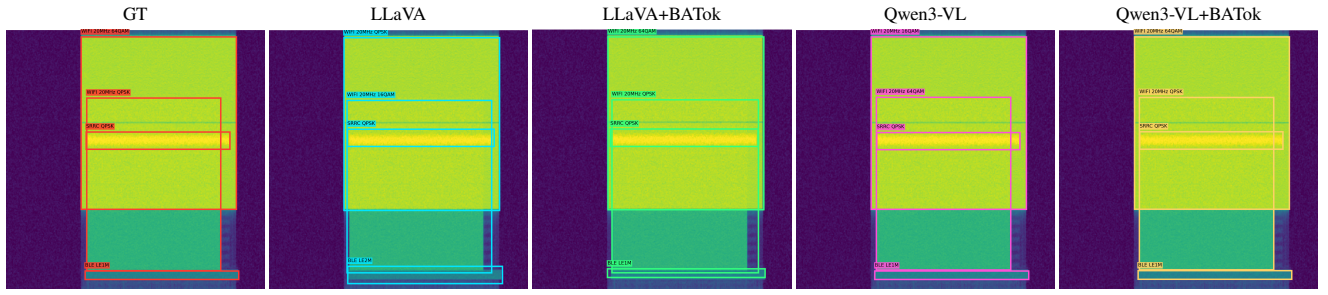


Figure S2: Detection in a crowded heterogeneous scene. **Question:** “Perform exhaustive detection and respond only with the requested tags.” **Ground truth:** [289, 25, 888, 691] (WiFi 20 MHz 64QAM), [304, 929, 897, 962] (BLE LE1M), [308, 392, 863, 459] (SRRRC QPSK), and [310, 261, 828, 928] (WiFi 20 MHz QPSK). Coordinates are normalized to [0, 1000]; the GT panel uses red boxes.

for Qwen3-VL-2B: in both figures, the image-only model labels the WiFi 20 MHz 64QAM transmission as 16QAM, whereas Qwen3-VL-2B+BATok recovers the correct class without degrading the corresponding box. The examples therefore illustrate the semantic contribution of raw I/Q observations alongside the geometric evidence supplied by the waterfall image.

S5.2 Language-Conditioned Grounding

The following cases vary the linguistic condition while keeping the same normalized box-output format. Figures S3 and S4 cover class and frequency constraints, followed by time and overlap relations.

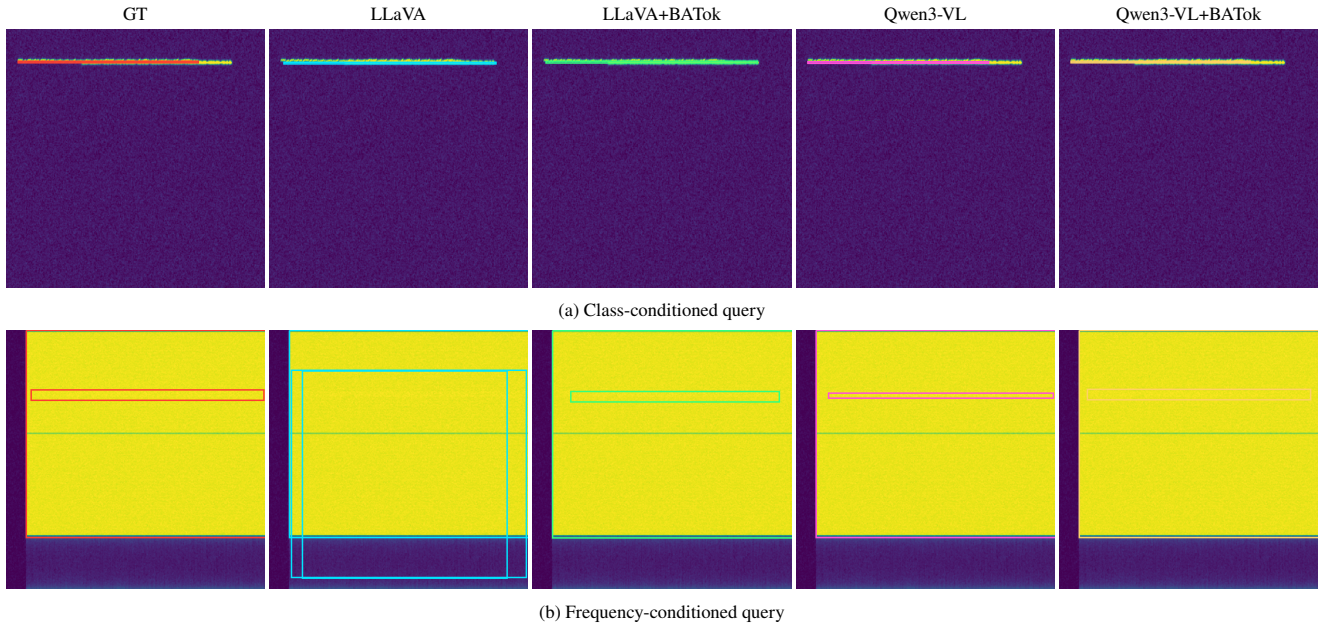


Figure S3: Grounding under class and frequency constraints. **(a) Question:** “Which regions correspond to the LoRa 250 kHz class?” **GT:** [46, 123, 741, 129]. **(b) Question:** “Find signal regions that overlap the frequency band 2455.015597–2457.837200 MHz.” **GT:** [77, 0, 1000, 800] and [96, 230, 994, 270]. Coordinates are normalized to [0, 1000]; GT panels use red boxes.

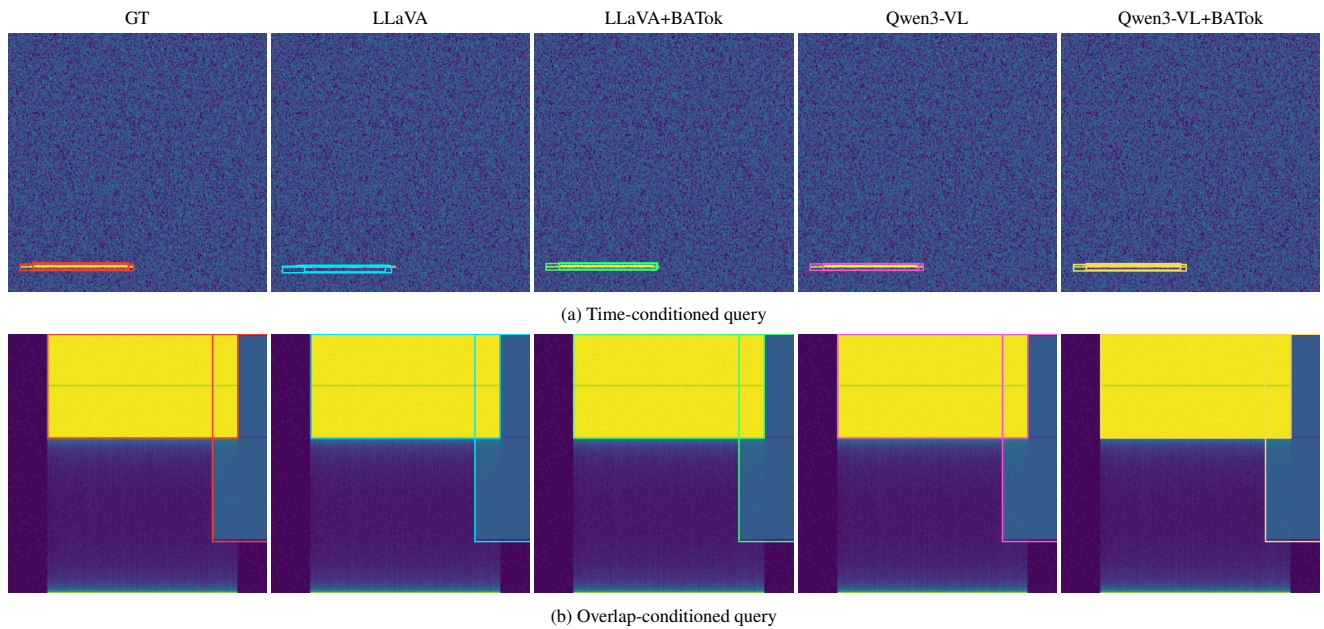


Figure S4: Grounding under time and overlap constraints. **(a) Question:** “Find signal regions that overlap the time interval 1.092–10.124 ms.” **GT:** [46, 891, 483, 916] and [96, 886, 463, 911]. **(b) Question:** “Find the signal regions involved in time–frequency overlap.” **GT:** [152, 0, 886, 400] and [788, 0, 1000, 800]. The overlap case is deliberately included because all compared variants succeed on it. Coordinates are normalized to [0, 1000]; GT panels use red boxes.

Taken together, the grounding examples show that the models can map class, frequency, time, and overlap conditions to localized signal instances under a common output format. The paired predictions further indicate that BATok primarily changes condition-dependent target selection while retaining the time–frequency geometry available from the image pathway. Figure S4 also shows that the additional signal pathway preserves correct predictions when the visual and linguistic evidence is already sufficient. These behaviors are consistent with the quantitative results: raw I/Q tokens complement waterfall images by supplying signal-level semantic evidence, while the visual pathway remains effective for spatial localization.